

Maximum Loadability of Islanded Microgrids With Renewable Energy Generation

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Abstract—A microgrid (MG) is a small-scale power system which is fed by constrained distributed generation (DG) units and its continuous operation is affected by the variability of available generation resources. In this paper, a global sensitivity analysis (GSA) method is proposed to evaluate the impact of variable energy resources on the maximum loadability of islanded MGs (IMGs). First, a probabilistic optimization problem is formulated to calculate the IMG load margin considering the droop characteristics of DG units and uncertainties of renewable generation, loads and distribution feeder parameters. Next, the global sensitivity (GS) is introduced that can identify the impact of independent and correlated variables on the IMG loadability. Last, the sparse polynomial chaos expansion method is used to obtain the probabilistic models for IMG load margins, and an efficient GSA method is proposed to calculate the GS of IMG loadability to prevailing variables. The probabilistic models are considered for IMG input variables and the impact of variable correlations on IMG loadability is analyzed. The proposed method for calculating the maximum loadability is tested using a 33-bus IMG and the results are compared with those of other GSA and local sensitivity analysis methods.

Index Terms—Microgrids, maximum loadability, global sensitivity analysis, polynomial chaos expansion.

NOMENCLATURE

Sets

Ω_B	Set of buses
Ω_{Gd}	Set of droop-controlled DG units
Ω_{Gpv}	Set of DG units at PV buses
Ω_s	Set of input random variables.

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Parameters

ω_0	Angular frequency at no-load
V_{i0}	Voltage magnitude of bus i at no-load
m_{pi}, n_{qi}	Active and reactive power droop gains
$P_{Gi,ub}, Q_{Gi,ub}$	Maximum active and reactive power of unit i
$P_{Gi,lb}, Q_{Gi,lb}$	Minimum active and reactive power of unit i
P_{Li}, Q_{Li}	Active and reactive power loads of bus i
P_{Li0}, Q_{Li0}	Nominal values of loads
α_{pi}, β_{qi}	Exponents of loads
h_{pi}, h_{qi}	Frequency sensitivity parameters
$P_{Li,b}, Q_{Li,b}$	Base active and reactive power loads
b_{pi}, b_{qi}	Active and reactive power load changes
k, c	Parameters of a Weibull distribution
r_{max}	Maximum solar irradiance
a, b	Parameters of a Beta distribution
Y_{ij}	Element of a bus admittance matrix
δ_{ij}	Phase angle of Y_{ij}
ε	Small positive number
p_i	Probability density function (PDF) of x_i
F_i	Cumulative distribution function (CDF) of x_i
ϕ, Φ	PDF and CDF of a normal variable
ϕ_n, Φ_n	PDF and CDF of a n -dimensional normal vector

μ	Mean vector
Σ	Covariance matrix
C	Copula function of correlated variables
n	Total number of input random variables
s	Number of variables in the subset y
N	Sample size of Monte Carlo simulation
N_c	Number of collocation points
$f_{1,...,lk}$	Item of ANOVA decomposition of function f
D, f_0	Variance and mean of f
$D_{1,...,lk}$	Variance of $f_{1,...,lk}$
S_y	Global sensitivity of y
a_i	Coefficient of polynomial chaos expansion (PCE)
H_i	i th order Hermite polynomial
m	Order of PCE.

Variables

ω	Angular frequency
λ	Load margin of IMG
V_i, θ_i	Voltage magnitude and angle of bus i

P_{Gi}, Q_{Gi}	Active and reactive power of a DG unit
P_{Ri}, Q_{Ri}	Active/reactive power of a renewable DG unit
P_{ij}	Active power flow from bus i to bus j
$V_{i,1}, V_{i,2}, V_{i,3}$	Items of voltage magnitude of bus i
x	Input random variable vector
x_i	i th input random variable
y, z	Subset of x and its complementary subset
$\bar{z}$	Variable vector of conditional probability distribution
γ, α, β	Normal variable vectors
$\xi_i, \hat{\xi}_i$	Normal variables.

I. INTRODUCTION

MICROGRID (MG) is operated either in grid-connected or islanded mode. In the grid-connected mode, the frequency and voltages of MGs are determined by the utility grid which is represented as an infinite bus. In islanded MGs (IMGs), the frequency and voltages are regulated by DG units governed by the respective droop characteristics. The IMG operation could enhance the reliability and the resilience of utility grid when the power system is facing outages. In the literature, various methods have been proposed for the planning and operation of IMGs [1]–[8].

The maximum loadability of transmission systems, which maintains the large-scale voltage stability, was studied extensively. For a distribution system (DS), the related work is quite limited since the DS loadability was traditionally managed by the connected main grid. Recently, the maximum IMG loadability has attracted additional attention since IMGs are fed in the island mode by small DG units which could enhance the loadability of IMG. The existing loadability approach evaluates the distance from the current operating point to the critical operating point for maintaining reliable and stable operations of IMGs. Reference [9] studied the maximum IMG loadability by calculating the equilibrium point using the Fischer-Burmeister-based method. Reference [10] evaluated the maximum loadability of droop-controlled IMGs using the optimal power flow method in which a complementary constraint was added to describe the capacity-limited DG unit operations. Reference [11] proposed a probabilistic algorithm to determine the optimal droop settings of IMGs. The algorithm adopted a hierarchical approach to expand the IMG load margin while satisfying the prevailing operating constraints.

IMGs can face various uncertainties including load and renewable energy forecast errors as well as random DS network outages. Accordingly, probabilistic methods are applied when evaluating the maximum IMG loadability in which the scenarios with minor impacts on IMG operations are disregarded. Therefore, a quick identification of critical variables could accelerate IMG loadability analyses and improve its controllability.

Global sensitivity analysis (GSA) is an effective method for quantifying the impact of IMG input variables on the MG performance [12]. Different from local sensitivity analysis (LSA), GSA considers system nonlinearities and interactions among pertinent variables. By decomposing the output

variance into segments attributed to a subset of inputs, we define the ratio of the segment to the total variance as the global sensitivity (GS) of input variables for evaluating IMG variables.

GSA was generally applied to artificial neural network [13] and hydrological modeling [14], with a limited applications to power systems. Recently, [15] introduced GSA to evaluate uncertainties in DS operations. Reference [16] compared GSA with LSA for identifying critical uncertainties in small signal stability analyses. However, GS was mostly calculated using a crude Monte Carlo simulation (MCS) method, which required large sample sizes to obtain accurate results, making GSA a time-consuming and expensive approach for large-scale problems. In [17] and [18], an improved GSA method was proposed to evaluate the impact of input variables on DS power flows. This method was improved by using sparse polynomial chaos expansion (PCE) to perform the high-dimensional model representation (HDMR) decomposition, thus presenting analytical GS solutions for independent variables. However, the HDMR decomposition is non-unique for correlated variables [19]. Besides, the correlation of variables, which affects the solution for uncertain variables, was not detailed in previous GSA applications.

In this paper, GSA is applied to IMGs with renewable energy generation. First, a probabilistic nonlinear optimization problem is established to calculate the load margin of IMGs considering DG droop characteristics and variabilities of renewable energy generation, loads and DS network parameters. Then, GSA is applied to IMG loadability with correlated input variables using conditional probability and Copula functions. Next, a surrogate model of load margin is established to calculate GS values, and an efficient GSA method is proposed to analyze the impact of input variables on the IMG load margin. The proposed GSA method is tested using a 33-bus IMG with various types of input variables, and the results are compared with those considering other sensitivity analysis methods to validate the effectiveness of the proposed GSA method.

II. MAXIMUM LOADABILITY ASSESSMENT OF IMG

In this section, the models are presented for DG units and loads. Then, probability models are developed for input variables. Finally, a probabilistic optimization problem is established to evaluate the maximum IMG loadability.

A. DG Model

There are three types of power system nodes, including PV, PQ and reference nodes, for balancing the total generation and load. In IMGs, DG units are responsible for regulating the frequency and voltages while satisfying load sharing. DG units in IMGs are usually modeled as droop, PV or PQ nodes.

In general, there are three droop control strategies: P - f / Q - V , P - V / Q - f and generalized droop, with the assumption of inductive, resistive and complex impedances, respectively. In this paper, the P - f / Q - V droop is adopted, which decouples active and reactive power controls and mimics the behavior of a synchronous generator in power systems. Moreover, a virtual

impedance control is applied to satisfy the inductive system requirement.

The proposed GSA method is applied to a general stochastic problem, which is not affected by the DG unit model. For the P - f / Q - V droop, active and reactive power of DG units are determined as

$$\begin{cases} P_{Gi} = (\omega_0 - \omega)/m_{pi} \\ Q_{Gi} = (V_{i0} - V_i)/n_{qi} \end{cases}, \quad i \in \Omega_{Gd} \quad (1)$$

For droop-controlled DG units, the output power follows the droop characteristic before it reaches the DG limits. However, beyond the active power generation limit $P_{Gi,ub}$, the DG unit is transformed into constant active power $P_{Gi,ub}$. Similarly, beyond the reactive power generation limit $Q_{Gi,ub}$, the DG unit is transformed into constant reactive power $Q_{Gi,ub}$ [10]. The droop-controlled capacity-limited DG units are described by the following constraints:

$$(P_{Gi,ub} - P_{Gi}) \geq 0, ((\omega_0 - \omega)/m_{pi} - P_{Gi}) \geq 0, \quad i \in \Omega_{Gd} \quad (2.1)$$

$$(P_{Gi,ub} - P_{Gi})((\omega_0 - \omega)/m_{pi} - P_{Gi}) = 0, \quad i \in \Omega_{Gd} \quad (2.2)$$

$$(Q_{Gi,ub} - Q_{Gi}) \geq 0, ((V_{i0} - V_i)/n_{qi} - Q_{Gi}) \geq 0, \quad i \in \Omega_{Gd} \quad (2.3)$$

$$(Q_{Gi,ub} - Q_{Gi})((V_{i0} - V_i)/n_{qi} - Q_{Gi}) = 0, \quad i \in \Omega_{Gd} \quad (2.4)$$

For DG units operating in a PV mode, we define $V_{i,1}$ as the desired voltage magnitude and use two non-negative variables $V_{i,2}$ and $V_{i,3}$ to describe the switching between PV and PQ modes, as stated in (3.1)-(3.4) [20]. If the reactive power output is between $Q_{Gi,lb}$ and $Q_{Gi,ub}$, (3.3) and (3.4) force $V_{i,2} = 0$ and $V_{i,3} = 0$, with a voltage magnitude that is equal to the desired $V_{i,1}$. When the reactive power output reaches its limit, the PV mode is switched to the PQ mode with constant reactive power. Specifically, if the reactive power output reaches its maximum, we have $Q_{Gi} = Q_{Gi,ub}$ and $Q_{Gi} > Q_{Gi,lb}$. Therefore, $V_{i,2} = 0$ according to (3.3) and $V_i = V_{i,1} - V_{i,3}$. Similarly, if the reactive power output reaches its minimum, we have $Q_{Gi} = Q_{Gi,lb}$ and $V_i = V_{i,1} + V_{i,2}$.

$$V_i = V_{i,1} + V_{i,2} - V_{i,3}, \quad V_{i,1}, V_{i,2}, V_{i,3} \geq 0, \quad i \in \Omega_{Gpv} \quad (3.1)$$

$$Q_{Gi,lb} \leq Q_{Gi} \leq Q_{Gi,ub}, \quad i \in \Omega_{Gpv} \quad (3.2)$$

$$(Q_{Gi} - Q_{Gi,lb})V_{i,2} = 0, \quad i \in \Omega_{Gpv} \quad (3.3)$$

$$(Q_{Gi,ub} - Q_{Gi})V_{i,3} = 0, \quad i \in \Omega_{Gpv} \quad (3.4)$$

B. Load Model

IMG loads, stated as a function of voltage magnitude and frequency, are given as

$$\begin{cases} P_{Li} = P_{Li0}(V_i/V_{i0})^{\alpha_{pi}}[1 + h_{pi}(\omega - \omega_0)] \\ Q_{Li} = Q_{Li0}(V_i/V_{i0})^{\beta_{qi}}[1 + h_{qi}(\omega - \omega_0)] \end{cases}, \quad i \in \Omega_B \quad (4)$$

In order to evaluate the maximum IMG loadability, b_{pi} and b_{qi} are used to describe incremental loads; thus, P_{Li0} and Q_{Li0} are expressed as

$$\begin{cases} P_{Li0} = P_{Li,b} + \lambda b_{pi} \\ Q_{Li0} = Q_{Li,b} + \lambda b_{qi} \end{cases}, \quad i \in \Omega_B \quad (5)$$

where $P_{Li,b}$ and $Q_{Li,b}$ are the base active and reactive loads, which represent the current operating point. λ is the system load margin which describes the distance between existing and critical operating points.

C. Modeling of Uncertainty

1) *Renewable-Based DG*: The probability distribution models for renewable energy are designed as follows. A Weibull model is adopted to describe the probability distribution of wind speed stated as

$$P(v) = \left(\frac{k}{c}\right)\left(\frac{v}{c}\right)^{k-1} \exp\left(-\left(\frac{v}{c}\right)^k\right), \quad v \geq 0 \quad (6)$$

A Beta model is used to describe the variability of solar irradiance for PV panels which is stated as

$$p(r) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \left(\frac{r}{r_{\max}}\right)^{a-1} \left(1 - \frac{r}{r_{\max}}\right)^{b-1} \quad (7)$$

2) *Load*: The variability of active power loads is described by a Normal distribution. The reactive power load will maintain a constant power factor.

3) *Parameter*: System parameters corresponding to generators, network, and loads are subject to forecast and measurement errors and system operating condition uncertainties. In this paper, parameter uncertainties are represented by a uniform distribution over a certain range [21].

4) *Correlation*: Copula functions are adopted to represent correlated variables [22].

D. Maximum Loadability Assessment Model

The load margin is an effective index to evaluate the maximum loadability of power systems. Considering the models of DG units, loads and their respective uncertainties, the following optimization problem is considered to calculate the IMG load margin:

$$\max \lambda \quad (8.1)$$

$$\text{subject to: } P_{Gi} + P_{Ri} - P_{Li} - V_i \sum V_j |Y_{ij}| \cos(\theta_{ij} - \delta_{ij}) = 0, \quad i \in \Omega_B \quad (8.2)$$

$$Q_{Gi} + Q_{Ri} - Q_{Li} - V_i \sum V_j |Y_{ij}| \sin(\theta_{ij} - \delta_{ij}) = 0, \quad i \in \Omega_B \quad (8.3)$$

$$V_{i,lb} \leq V_i \leq V_{i,ub}, \quad i \in \Omega_B \quad (8.4)$$

$$P_{ij,lb} \leq P_{ij} \leq P_{ij,ub}, \quad (i, j) \in \Omega_B \quad (8.5)$$

$$P_{Gi,lb} \leq P_{Gi} \leq P_{Gi,ub}, \quad i \in \{\Omega_{Gd}, \Omega_{Gpv}\} \quad (8.6)$$

$$Q_{Gi,lb} \leq Q_{Gi} \leq Q_{Gi,ub}, \quad i \in \{\Omega_{Gd}, \Omega_{Gpv}\} \quad (8.7)$$

$$(\omega_0 - \omega)/m_{pi} - P_{Gi} \geq 0, \quad i \in \Omega_{Gd} \quad (8.8)$$

$$(V_{i0} - V_i)/n_{qi} - Q_{Gi} \geq 0, \quad i \in \Omega_{Gd} \quad (8.9)$$

$$(P_{Gi,ub} - P_{Gi})((\omega_0 - \omega)/m_{pi} - P_{Gi}) \leq \varepsilon, \quad i \in \Omega_{Gd} \quad (8.10)$$

$$(Q_{Gi,ub} - Q_{Gi})((V_{i0} - V_i)/n_{qi} - Q_{Gi}) \leq \varepsilon, \quad i \in \Omega_{Gd} \quad (8.11)$$

$$(Q_{Gi} - Q_{Gi,lb})V_{i,2} \leq \varepsilon, (Q_{Gi,ub} - Q_{Gi})V_{i,3} \leq \varepsilon, \quad i \in \Omega_{Gpv} \quad (8.12)$$

$$V_i = V_{i,1} + V_{i,2} - V_{i,3}, \quad i \in \Omega_{Gpv} \quad (8.13)$$

$$V_{i,1} \geq 0, V_{i,2} \geq 0, V_{i,3} \geq 0, \quad i \in \Omega_{Gpv} \quad (8.14)$$

$$F_i(x) = \int_{-\infty}^x p_i(t) dt, \quad i \in \Omega_s \quad (8.15)$$

$$F(x_1, \dots, x_n) = C(F_1(x_1), \dots, F_n(x_n)) \quad (8.16)$$

In the above formulation, the objective function is to maximize the IMG load margin. Constraints (8.2) and (8.3) represent power balance. Constraints (8.4)-(8.7) are limits on

voltage magnitudes, feeder power flows and DG power outputs. Constraints (8.8)-(8.11) describe the behavior of droop-controlled DG units. Constraints (8.12)-(8.14) describe the switching between PV and PQ modes. (8.15) is the probability distribution of variable x_i and (8.16) is the Copula function of correlated variables. Besides, (2.2), (2.4), (3.3) and (3.4) are transformed to inequations in (8.10)-(8.12), and a small positive number ($\varepsilon = 10^{-7}$) is introduced to guarantee a strictly feasible region of the optimization problem [20].

III. GLOBAL SENSITIVITY ANALYSIS

In this section, GSA is developed for independent and dependent random variables, and the GS calculation is discussed.

A. GSA With Independent Variables

Assume $x = (x_1, \dots, x_n)$ are uniformly distributed in an n -dimensional unit hypercube. Let $Y = f(x_1, \dots, x_n)$ be a function with n input variables represented as

$$f(x) = f_0 + \sum_{i=1}^n f_i(x_i) + \sum_{i<j}^n f_{ij}(x_i, x_j) + \dots + f_{1\dots n}(x_1, \dots, x_n) \quad (9)$$

Which is defined as the analysis of variation (ANOVA) decomposition of f if the following condition is satisfied:

$$\int f_{l_1\dots l_k}(x_{l_1}, \dots, x_{l_k}) dx_t = 0, t = l_1, \dots, l_k \quad (10)$$

The ANOVA decomposition is unique only if variables are independent. Moreover, using (10) we have,

$$D = \sum_{i=1}^n D_i + \sum_{i<j}^n D_{ij} + \dots + D_{1\dots n} \quad (11)$$

where D is the total variance of f and $D_{l_1\dots l_k}$ is the variance of $f_{l_1\dots l_k}(x_{l_1}, \dots, x_{l_k})$, which are calculated as

$$\begin{cases} D = \int f^2 dx - f_0^2 \\ D_{l_1\dots l_k} = \int f_{l_1\dots l_k}^2 dx_{l_1} \dots dx_{l_k} \end{cases} \quad (12)$$

Consider an arbitrary set of s variables $y = (x_{i_1}, \dots, x_{i_s})$, $1 \leq s < n$, and a complementary subset $z = (x_{j_1}, \dots, x_{j_{n-s}})$, such that $x = (y, z)$. The variance corresponding to y is defined as

$$D_y = \sum_{k=1}^s \sum_{(l_1, \dots, l_k) \in I_s} D_{l_1\dots l_k}, I_s = i_1, \dots, i_s. \quad (13)$$

The variance-based GS index S_y is defined to evaluate the impact of input variables y on the system output:

$$S_y = \frac{D_y}{D}. \quad (14)$$

B. GS Calculation for Independent Variables

For independent uniformly distributed variables, the subset's variance D_y is stated as [12]:

$$D_y = \int f(x)f(y, z') dx dz' - f_0^2 \quad (15)$$

Therefore, S_y can be estimated by MCS as

$$\begin{cases} S_y = \frac{1}{D} \left[\frac{1}{N} \sum_{k=1}^N (f(y(k), z(k))f(y(k), z'(k))) - f_0^2 \right] \\ f_0 = \frac{1}{N} \sum_{k=1}^N f(y(k), z(k)) \\ D = \frac{1}{N} \sum_{k=1}^N (f(y(k), z(k)))^2 - f_0^2 \end{cases} \quad (16)$$

where N is the MCS sample size, k indicates the sample number for the variable vector, and $z_{(k)}$ and $z'_{(k)}$ represent two sample vectors of z .

C. GSA With Dependent Variables

When variables are correlated, the tight correspondence between functional and variance decompositions is lost as the variance cannot be decomposed into a sum of squares as given in (11) [23]. For correlated variables, the total variance of $f(x)$ is decomposed as

$$D = D_y[E_z[f(y, \bar{z})]] + E_y[D_z[f(y, \bar{z})]] \quad (17)$$

where

$$\begin{cases} E_z[f(y, \bar{z})] = \int_{R^{n-s}} f(y, \bar{z}) p(y, \bar{z}|y) d\bar{z} \\ D_y[E_z[f(y, \bar{z})]] = \int_{R^s} [E_z(f(y, \bar{z}))]^2 p(y) dy - f_0^2 \\ f_0 = \int_{R^n} f(y, z) p(y, z) dy dz \end{cases} \quad (18)$$

where z is a random vector produced from the joint PDF $p(y, z)$, and $\bar{z}$ is a random vector produced from the conditional probability distribution $p(y, \bar{z}|y)$.

Based on the definition of variance-based sensitivity, the GS of subset y is given as

$$S_y = \frac{D_y[E_z[f(y, \bar{z})]]}{D} \quad (19)$$

Substituting (18) into (19), the full expression of S_y is

$$S_y = \frac{1}{D} \left[\int_{R^s} \left[\int_{R^{n-s}} f(y, \bar{z}) p(y, \bar{z}|y) d\bar{z} \right]^2 p(y) dy - f_0^2 \right] \quad (20)$$

MCS can be used to estimate S_y , but the direct application of (20) requires costly computations of a double loop [24]. Therefore, a more efficient derivation [19] is stated as:

$$S_y = \frac{1}{D} \left[\int_{R^s} p(y) dy \left[\int_{R^{n-s}} f(y, \bar{z}) p(y, \bar{z}|y) d\bar{z} \times \int_{R^{n-s}} f(y, \bar{z}') p(y, \bar{z}'|y) d\bar{z}' - f_0^2 \right] \right]. \quad (21)$$

D. GS Calculation for Correlated Normal Variables

For n normal variables $\gamma = (\xi_1, \dots, \xi_n)$, consider a subset $\alpha = (\xi_{i_1}, \dots, \xi_{i_s})$, $1 \leq s < n$ and its complementary subset β . The mean vector and the covariance matrix of γ are partitioned as

$$\begin{cases} \mu_\gamma = [\mu_\alpha \quad \mu_\beta]^T \\ \Sigma_\gamma = \begin{bmatrix} \Sigma_\alpha & \Sigma_{\alpha\beta} \\ \Sigma_{\beta\alpha} & \Sigma_\beta \end{bmatrix} \end{cases} \quad (22)$$

The PDF of the n -dimensional normal variable vector γ is

$$\phi_n(\gamma) = \frac{1}{\sqrt{(2\pi)^n |\Sigma_\gamma|}} \exp\left(-\frac{1}{2}(\gamma - \mu_r)^T \Sigma_\gamma^{-1} (\gamma - \mu_r)\right) \quad (23)$$

For correlated normal variables, the conditional distribution $\phi_{n-s}(\alpha, \bar{\beta}|\alpha)$ is also a normal distribution with the mean vector $\mu_{\beta c}$ and the covariance matrix $\Sigma_{\beta c}$ stated as

$$\begin{cases} \mu_{\beta c} = \mu_\beta + \Sigma_{\beta\alpha} \Sigma_\alpha^{-1} (\alpha - \mu_\alpha) \\ \Sigma_{\beta c} = \Sigma_\beta - \Sigma_{\beta\alpha} \Sigma_\alpha^{-1} \Sigma_{\alpha\beta} \end{cases} \quad (24)$$

Therefore, for correlated normal variables, (21) becomes

$$S_\alpha = \frac{1}{D} \left[\int_{R^s} \phi_s(\alpha) d\alpha \left[\int_{R^{n-s}} f(\alpha, \bar{\beta}) \phi_{n-s}(\alpha, \bar{\beta}|\alpha) d\bar{\beta} \times \int_{R^{n-s}} f(\alpha, \bar{\beta}') \phi_{n-s}(\alpha, \bar{\beta}'|\alpha) d\bar{\beta}' \right] - f_0^2 \right]. \quad (25)$$

E. GS Calculation for Correlated Non-Normal Variables

Let the marginal probability distribution of variable x_i be F_i ($i = 1, \dots, n$). The Gaussian Copula is used to establish the joint probability distribution of $x_1, \dots, x_n$ with the following formula:

$$C(F_1(x_1), \dots, F_n(x_n)) = \Phi_n\left(\Phi^{-1}(F_1(x_1)), \dots, \Phi^{-1}(F_n(x_n)); \Sigma\right) \quad (26)$$

The PDF of the Gaussian Copula satisfies the following

$$p(F_1(x_1), \dots, F_n(x_n)) = \frac{\phi_n(\Phi^{-1}(F_1(x_1)), \dots, \Phi^{-1}(F_n(x_n)); \Sigma)}{\prod_{i=1}^n \phi(\Phi^{-1}(F_i(x_i)))} \quad (27)$$

According to the transformation law of probabilities (27), the GS for correlated non-normally distributed variables is presented as [19]

$$S_y = \frac{1}{D} \left[\int_{R^s} \phi_s(\alpha) d\alpha \left[\int_{R^{n-s}} f(H_s(\alpha), H_{n-s}(\bar{\beta})) \phi_{n-s}(\alpha, \bar{\beta}|\alpha) d\bar{\beta} \times \int_{R^{n-s}} f(H_s(\alpha), H_{n-s}(\bar{\beta}')) \phi_{n-s}(\alpha, \bar{\beta}'|\alpha) d\bar{\beta}' \right] - f_0^2 \right] \quad (28)$$

where

$$\begin{cases} H_s(\alpha) = (F_{i_1}^{-1}(\Phi(\xi_{i_1})), \dots, F_{i_s}^{-1}(\Phi(\xi_{i_s}))) \\ H_{n-s}(\bar{\beta}) = (F_{j_1}^{-1}(\Phi(\xi_{j_1})), \dots, F_{j_{n-s}}^{-1}(\Phi(\xi_{j_{n-s}}))) \end{cases} \quad (29)$$

Based on (28), S_y is estimated by MCS as

$$S_y = \frac{1}{D} \left[\frac{1}{N} \sum_{k=1}^N (f(y(k), z(k)) f(y(k), \bar{z}(k))) - f_0^2 \right] \quad (30)$$

The application of (30) requires: 1) samples of joint and conditional probability distributions, 2) evaluation of system output based on inputs. The procedure to perform GSA for the maximum IMG loadability is given in the following section.

IV. GSA FOR MAXIMUM IMG LOADABILITY

MCS is usually used to perform GSA which requires many samples to obtain accurate results, which limits the practical GSA applications. In this paper, an efficient GSA method is proposed to analyze the impacts of variables on the loadability of IMGs. Instead of directly calculating GSs with MCS, a surrogate model for the load margin is first established with PCE, and then GS is calculated using the surrogate model.

A. Polynomial Chaos Expansion

PCE is used to represent the random model output by a set of coefficients in polynomial chaos bases. For n independent standard normal variables $\hat{\xi}_1, \dots, \hat{\xi}_n$, PCE is stated as

$$Y = a_0 + \sum_{i_1=1}^n a_{i_1} H_1(\hat{\xi}_{i_1}) + \sum_{i_1=1}^n \sum_{i_2=1}^{i_1} a_{i_1 i_2} H_2(\hat{\xi}_{i_1}, \hat{\xi}_{i_2}) + \sum_{i_1=1}^n \sum_{i_2=1}^{i_1} \sum_{i_3=1}^{i_2} a_{i_1 i_2 i_3} H_3(\hat{\xi}_{i_1}, \hat{\xi}_{i_2}, \hat{\xi}_{i_3}) + \dots \quad (31)$$

where $a_0, a_{i_1}, a_{i_1 i_2}$ and $a_{i_1 i_2 i_3}$ are PCE coefficients, and H_i is the i th order Hermite polynomial ($i = 1, \dots, m$).

The number of coefficients in (31) is $(n+m)!/(n!m!)$, which grows by increasing n or m . Moreover, a sufficient number of samples, like 2-3 times the number of coefficients, is used to estimate these coefficients, making PCE computationally intractable for large uncertain problems. Sparse PCE (SPCE) was proposed by Blatman and Sudret to address the PCE problem when the output is dominated by lower order interactions between variables. The main idea of SPCE is to apply a least angle regression-based truncation scheme to detect significant PCE coefficients and retain a smaller number of PCE terms. The theory of truncation scheme is presented in [25] as we use the UQLab toolbox to construct the SPCE model in this paper [26].

B. Procedure of GSA

The probabilistic nonlinear optimization problem (7) is expressed in a compact form as $Y = f(x)$, where x are n random variables, and Y is the IMG load margin. The GSA procedure for evaluating the impact of variables on the load margin is given as follows.

Step 1: Obtain the IMG parameters and the probability models for random variables

Step 2: Select a set of variables, defined as y , from n input variables x

Step 3: Establish surrogate models of load margins

- Select N_c samples of independent standard normal variables as collocation points of input variables
- Transform collocation points into samples with a joint distribution $p(y, z)$ or a conditional distribution $p(y, \bar{z}|y)$
- Solve the deterministic optimization problem (8) and obtain load margin samples
- Calculate SPCE coefficients based on the collocation points and load margin samples
- Establish the surrogate models of $f(y, z)$ or $f(y, \bar{z})$

Step 4: Obtain surrogate model output using MCS

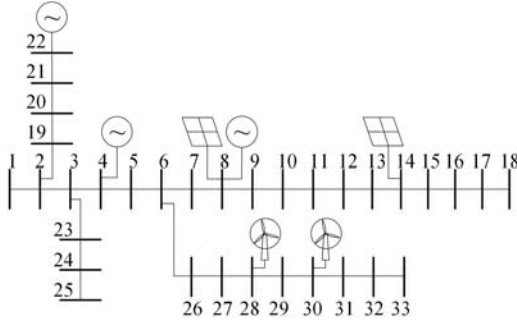


Fig. 1. 33-bus IMG test system.

In order to improve the MCS efficiency, the Sobol sequence is used to obtain the surrogate model output using the following steps:

- i. Generate $2n$ -dimensional Sobol sequences with a sample size of N
- ii. Transform the first n -dimensional Sobol sequences into samples of standard normal variables, and obtain the surrogate model output of $f(y, z)$ using the samples
- iii. Combine the first s -dimensional with the last $n-s$ -dimensional Sobol sequences, transform them into samples of standard normal variables, and obtain the surrogate model output of $f(y, \bar{z})$ using the samples

Step 5: Calculate the GS using (30) and assess the importance of variables y .

V. CASE STUDIES

In this section, the proposed GSA method is tested using a 33-bus IMG. The variabilities of loads, feeder parameters and renewable power are modeled as random variables, and the correlation of renewable power outputs is considered. The proposed method is compared with other GSA and LSA methods for evaluating the impacts of variables. Programs are developed with MATLAB 8.1 on a PC with Intel Core i7 3.6GHz CPU and 8GB of RAM. Sobol sequences are used to generate samples of random variables. The OPTI toolbox [27] is used to solve the nonlinear optimization problem (8). The UQLab toolbox is used to calculate SPCE coefficients.

A. IMG System Description

Fig. 1 shows the diagram of the 33-bus IMG system. The feeder parameters and nominal load power of the system are given in [28]. Three dispatchable DG units and four renewable-based DG units are allocated to feed the IMG system, and the parameters of DG units are given in Tables I and II [11]. The load parameters are designed as $\alpha_p = 0.92$, $\beta_q = 4.04$, $h_p = 0.8$ and $h_q = -2.0$ [29]. The upper and lower limits of voltage magnitude are 1.05 and 0.95 p.u. The upper and lower limits of the system frequency are 1.005 and 0.995 p.u.

The probability models of input random variables are given in Table III. The probability distribution models of wind speed and solar irradiance are described by Weibull and Beta distributions, respectively [30]. The active power load has a normal

TABLE I
PARAMETERS OF DROOP-CONTROLLED DG UNITS ($S_{\text{base}} = 1\text{MVA}$)

Bus	$P_{G,ub}$ (p.u.)	$Q_{G,ub}$ (p.u.)	m_p	n_q	V_0	ω_0
4	3	1.5	7.29E-4	1.65E-5	1.0009	1.016
8	0.5	1	4.03E-3	6.17E-2	1.0008	1.016
22	1.5	1	1.22E-3	3.21E-2	1.0009	1.040

TABLE II
PARAMETERS OF RENEWABLE-BASED DG UNITS ($S_{\text{base}} = 1\text{MVA}$)

Bus	Type	Capacity (p.u.)	Power factor
8	PV	0.75	0.9
14	PV	0.75	0.9
28	Wind	0.75	0.9
30	Wind	0.75	0.9

TABLE III
MODELS OF INPUT RANDOM VARIABLES

Index	Variable	Model	Parameter
1-2	Wind	Weibull	$c=7.41, k=2.06$
3-4	PV	Beta	$a=2.06, b=2.5$
5-36	Load	Normal	$\sigma/\mu=0.2$
37-73	Feeder parameter	Uniform	$lb=0.9, ub=1.1$

distribution with a standard deviation which is 20% of its mean. The reactance of a feeder l is in $[0.9R_l, 1.1R_l]$, where R_l is the base reactance value. The resistance is modeled to maintain a constant R/X . Moreover, the power outputs of the same type of renewable-based DG units have a rank correlation coefficient of 0.5 for the Gaussian Copula. There are 73 variables in IMG including independent and correlated variables.

B. Surrogate Model

In this section, the surrogate models of load margins are established by PCE and SPCE. The 2nd-order PCE with accurate results in power system simulations [30] is used in our simulations. Note that higher order PCEs can also be used. There are 73 variables in the test system; thus the number of coefficients of the 2nd-order PCE is 2,775. Accordingly, 8,000 collocation points are used to estimate the coefficients using a least square method [17]. The 2nd-order PCE is also applied and 1,000 collocation points are used to estimate the SPCE coefficients. For comparison, MCS with a sample size of 50,000 is used to solve the probabilistic nonlinear optimization problem (8).

The probability distributions of load margins are given in Fig. 2. The deterministic load margin is 0.49. However, if input random variables are considered, based on SPCE results, the probability that the load margin is larger than 0.7 is 11.4% and the probability that the load margin is smaller than 0.3 is 16.6%. Therefore, input variables affect the variability of load margins significantly. Moreover, CDFs of load margins obtained by SPCE, PCE and MCS are nearly the same, which validates the accuracy of surrogate models.

The coefficients of PCE and SPCE are given in Fig. 3. Although there are 2,775 coefficients in PCE and only 140 non-zero coefficients in SPCE, the coefficients with absolute values larger than 0.01 are almost the same, making the

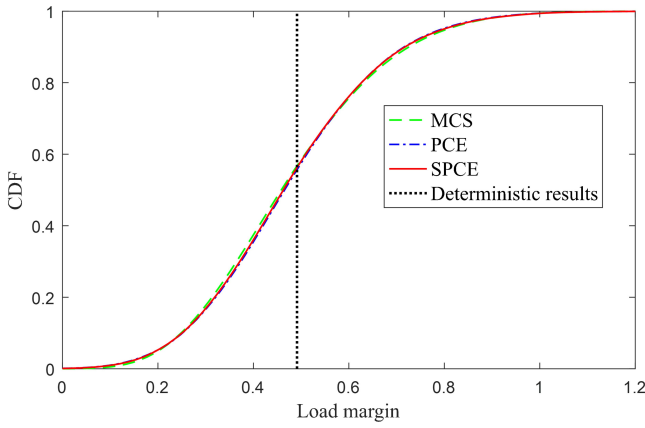


Fig. 2. CDFs of load margin.

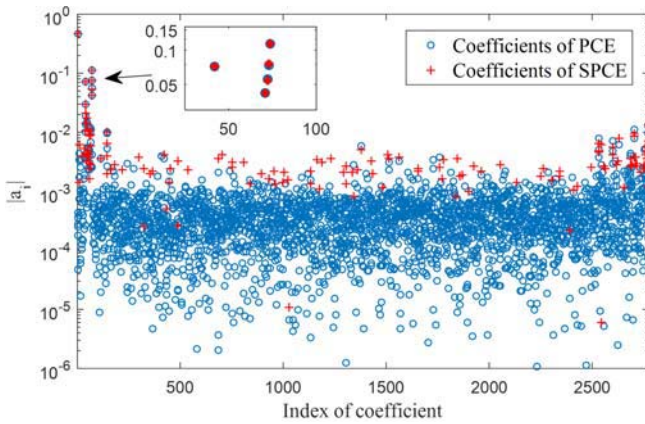


Fig. 3. Coefficients of PCE and SPCE.

results of PCE and SPCE consistent. Compared with PCE, SPCE obtains accurate results with much fewer coefficients for high-dimensional and uncertain problems and thus fewer collocation points are needed to estimate coefficients.

C. Sensitivity Analysis

In this section, the proposed method is used to calculate sensitivities of random variables with respect to IMG load margins. The performances of GSA-MCS, GSA-PCE and GSA-SPCE are compared. The GS of wind power at bus 28 is given in Fig. 4. The three methods obtain similar GSs with large sample sizes. The result of GSA-PCE is closer to that of GSA-MCS, since full polynomial chaos bases are retained in PCE while only significant polynomial chaos bases are retained in SPCE. Nevertheless, GSA-SPCE also obtains relatively accurate GSs. It is found that GSA obtains stable results when more than 10,000 samples are considered. Therefore, GSA suffers from heavy computational burdens if GSs are calculated based on the original problem. The sample size and computation time of corresponding methods are given in Table IV. Although GSA-PCE and GSA-SPCE require additional time for establishing surrogate models, they are much more efficient than GSA-MCS since GSs are calculated based on surrogate models. GSA-SPCE is more efficient

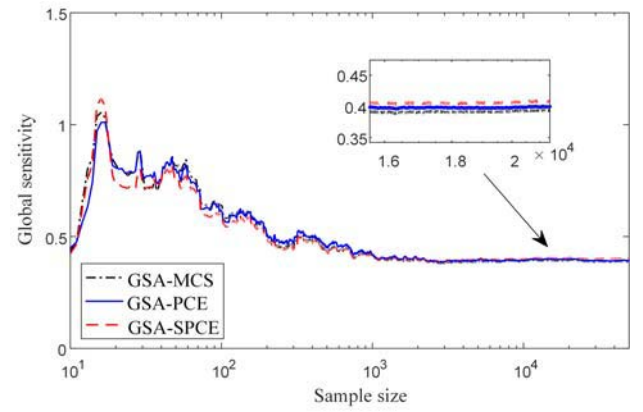


Fig. 4. Relation between GS and sample size.

TABLE IV
SAMPLE SIZES AND COMPUTATION TIME OF DIFFERENT
GSA METHODS

		GSA-MCS	GSA-PCE	GSA-SPCE
Establishing surrogate models	Sample size	-	8000	1000
	Time (s)	-	684.7	89.8
Calculating GSs	Sample size	50000	50000	50000
	Time (s)	8796.4	1.5	0.6
Total computation time (s)		8796.4	686.2	90.4

than GSA-PCE because fewer samples are used to estimate surrogate model coefficients.

The GS of each input variable is given in Fig. 5. There are 73 variables in total and only those with GSs larger than 0.01 are presented. In Fig. 5, the results obtained by GSA-SPCE and GSA-PCE are nearly the same for all variables. Moreover, only a small set of variables have relatively large GSs, and most of the variables have minor impacts on load margins. When the renewable generation output is assumed to be deterministic, the 33rd input variable, which is the load demand at bus 30, accounts for most of the load margin variability with a GS that is 0.656 for GSA-PCE and 0.659 for GSA-SPCE. When the renewable power uncertainty is considered, the load impact decreases. The wind power is the most influencing factor, followed by the load at bus 30 and PV power. The GSs of feeder parameters are smaller than 0.01.

GSA is also compared with the commonly used LSA [15] in assessing the importance of variables and 6 variables with largest sensitivities are given in Table V. The critical variables among independent variables identified by LSA and GSA are the same, but the rankings of variables are different. This is because local sensitivities (LSs) are based on the partial derivative of the output with respect to the input while GSs consider nonlinear systems and apply the probability models of variables. Hence GSA is more accurate for evaluating the impact of variables. Besides, LSA cannot be used for correlated variables and LSs do not provide any credible information on the segment of output uncertainty which is attributed to a subset of inputs.

Finally, the impact of the correlations of input variables is investigated. The rank correlation of Gaussian Copula varies between -1 and 1 , and GSs for various types of uncertainties

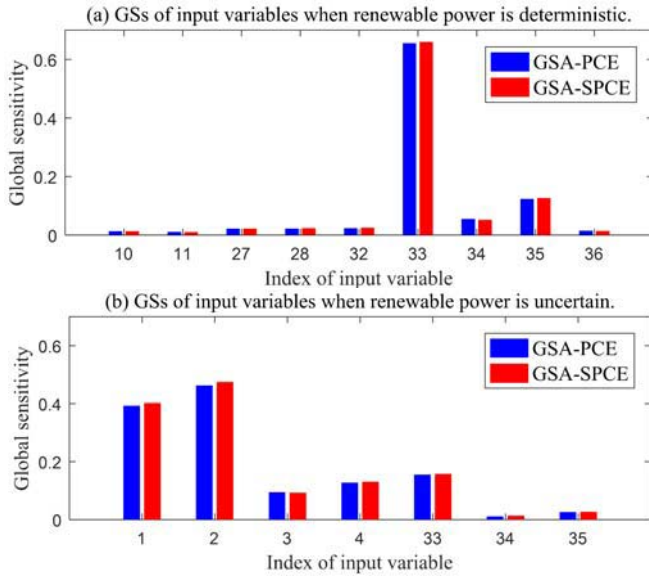


Fig. 5. GSs of random variables.

TABLE V
SENSITIVITIES OF VARIABLES OBTAINED BY LSA AND GSA

Ranking	LSA		GSA-SPCE			
	Variable	LS	Variable	GS	Variable	GS
1	2	0.225	2	0.312	2	0.474
2	33	0.170	1	0.213	1	0.403
3	1	0.166	33	0.210	33	0.157
4	35	0.075	4	0.087	4	0.131
5	3	0.057	35	0.034	3	0.093
6	4	0.055	3	0.032	35	0.026

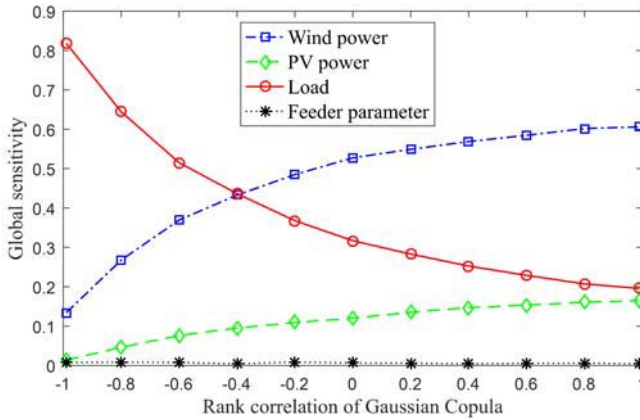


Fig. 6. GS of different types of correlated uncertainties.

are given in Fig. 6. The GS of wind and PV power increases and the GS of loads decreases as we increase the correlation coefficient of renewable power. The GS of feeder parameters remains small in all simulation cases.

The GS of wind power with various correlations are given in Fig. 7. For independent wind power, the sum of the GS of each wind power is almost equal to the GS of the total wind power, which indicates that the wind power interactions nearly have no impact on load margins. For positively or negatively

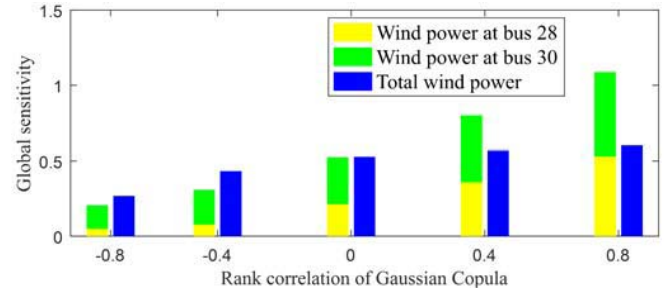


Fig. 7. GS of each wind power and the total wind power.

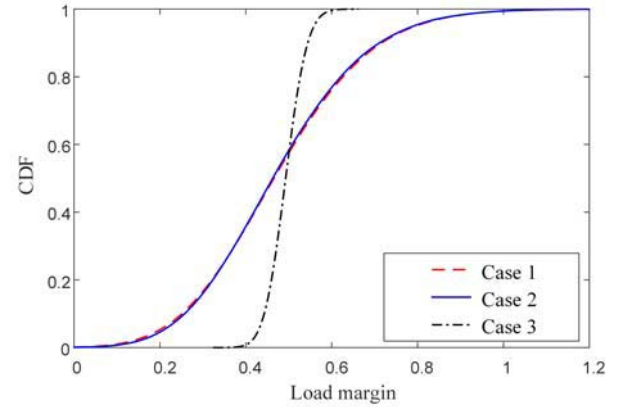


Fig. 8. Cumulative distributions of load margin when different variables are considered.

correlated wind power, the sum of the GS of each wind power is larger or smaller than the GS of total wind power. This is because of the variable attributes and its correlations with other variables are considered for calculating GS. The sum of the GS of each wind power will similarly consider the impact of correlations.

D. Maximum Loadability Analysis

In order to validate the GSA results, we consider the following three cases and evaluate the probabilistic value of the maximum IMG loadability.

Case 1: All 73 input random variables are considered.

Case 2: 7 variables with GSs larger than 0.01 are considered.

Case 3: 66 variables with a GS smaller than 0.01 are considered.

Fig. 8 shows that if we merely consider the 7 variables with large GSs, the load margin CDFs will almost be the same. However, if we only consider the 66 variables with small GSs, the load margin CDFs will be quite different although most variables are considered. This result indicates that most variables have a minute impact on the load margin, and the proposed method accurately identifies the most effective variables.

The bus voltage magnitudes and droop-controlled DG unit outputs at critical operating points are given in Figs. 9 and 10, respectively. The minimum voltage magnitude is 0.95 which reaches the lower limit. The maximum voltage magnitude is 1.03 at bus 22 which is smaller than the upper limit. For some buses, voltage magnitude variations are different for cases 1

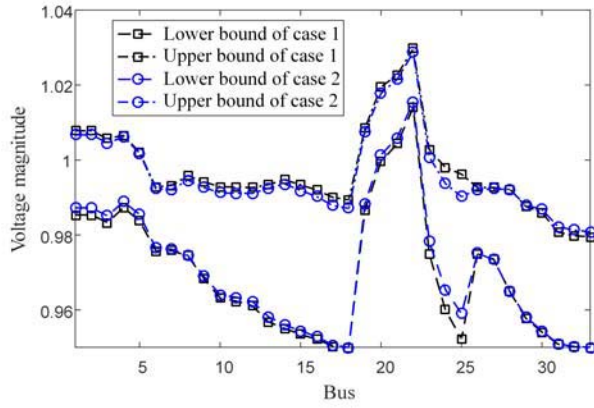


Fig. 9. Bus voltage magnitudes at critical operating points when different variables are considered.

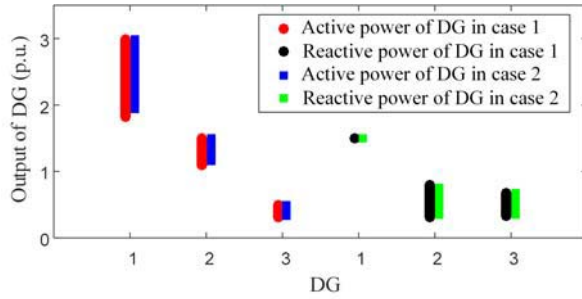


Fig. 10. DG unit outputs at critical operating points when different variables are considered.

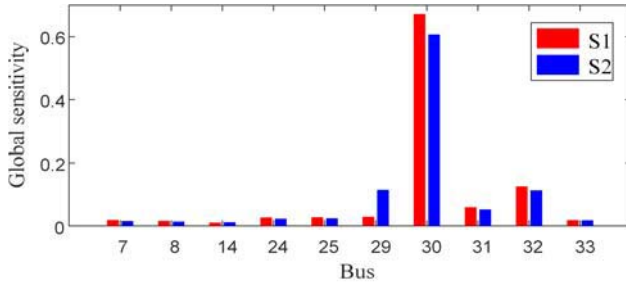


Fig. 11. GS of load demand at each bus.

and 2, which indicates that although some variables have an impact on voltage magnitudes, they do not affect the load margin. The DG unit outputs in cases 1 and 2 are similar. The reactive power of unit 1 at critical operating points is always equal to its maximum, and the active or reactive power of other units varies within certain ranges.

E. Influence of PHEV

As the number of plug-in hybrid electric vehicles (PHEVs) increases rapidly, the charging demand has become an important part of the electricity load. In the following, we use the PHEV model in [31] to evaluate the load demand variability of charging stations. The method establishes a single PHEV charging demand model and employs queuing theory to describe the behavior of multiple PHEVs. Finally, the probability model of the total demand of a PHEV

TABLE VI
LOAD DEMANDS OF PHEV CHARGING STATIONS

Bus	Weibull distribution		Mean power (MW)	Power factor
	Scale parameter	Shape parameter		
22	0.192329	1.87103	0.17076	0.95
29	0.192329	1.87103	0.17076	0.95

charging station is obtained with the maximum likelihood estimation. We consider two cases to analyze the GS of PHEVs.

S1: All loads are described by normal distributions.

S2: The loads at buses 22 and 29 are replaced by PHEV charging stations with models shown in Table VI.

The GS of each variable is calculated and load demands with GSs larger than 0.01 are given in Fig. 11. When load demands are described by normal distributions, the load demand at bus 30 accounts for most of the load margin variability with a GS equal to 0.672. When the integration of PHEV charging stations is considered, the GS of load demand at bus 30 decreases and the GS of load demand at bus 29 increases. In addition, the GS of the PHEV charging demand at bus 22 remains small, although the PHEV charging demand at bus 22 has the same probability model as that at bus 29. Therefore, compared with load demands described by normal distributions, the load demand of PHEV charging stations in this case has a higher impact on the maximum IMG loadability. The influence of charging demand will decrease by selecting suitable locations for PHEV charging stations.

VI. CONCLUSION

In this paper, a GSA method is proposed to evaluate the impact of variables on the maximum IMG loadability. Considering MG characteristics operating in the islanded mode and the probability distribution models of uncertainties, a probabilistic optimization problem is established to evaluate IMG load margins. Then, GSs are presented to evaluate the importance of independent and correlated variables. In order to improve the efficiency of GSA, SPCE is used to establish probability models of load margins, and GSs are calculated based on surrogate models.

The proposed method is tested using a 33-bus IMG system which is compared with other SA methods. The simulation results indicate that SPCE overcomes the PCE deficiency in high-dimensional problems by reducing the number of polynomial chaos bases, thus much fewer coefficients are estimated in establishing surrogate models. Compared with LSs, GSs consider the power system nonlinearity and describe the impact of variable dependence on the output. GSA-SPCE obtains accurate GSs and is much more efficient than GSA-MCS and GSA-PCE. For IMGs with various uncertainties, accurate identification of the most influencing variables will reduce the number of variables in IMG analyses and help manage the impact of variabilities with proper control strategies, such as the installation of energy storage systems.

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